MACHINE LEARNING
INTRODUCTION AND APPLICATIONS

1. Introduction to Machine Learning (ML)

Definition: Machine Learning is a subset of Artificial Intelligence (AI) that enables machines to learn from data and make decisions or predictions without being explicitly programmed.

Key Points to Remember:
- ML is based on data and algorithins.
- The system learns patterns from data and improves over time.
- It's used when explicit programming is difficult or impossible.

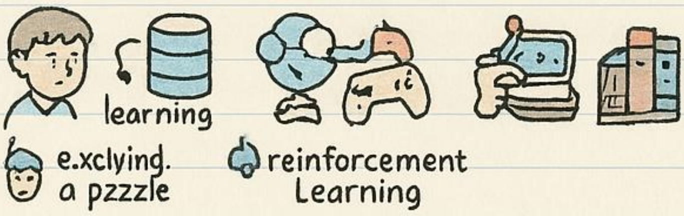
Real-Life Analogy:

Supervised Learning: Like a student learning with an answer key.

Unsprvised Learning: Like solving a pazzle without knowing picture.

Types of Machine Learning:

Type	Description	Example
Supervised Learning	Learns from labeled data Learns from unlabeled dta	Predicting house prices Customer segmentation
Unoupervised Learning	Learns by trial and error	Self-driving cars, games



Reinforcement Learning
Like learning plag a video game by getting or penalties.

2. Applications of Machine Learning

Machine Learning is used in almost every industry. Below are common applications, grouped for easy recall.

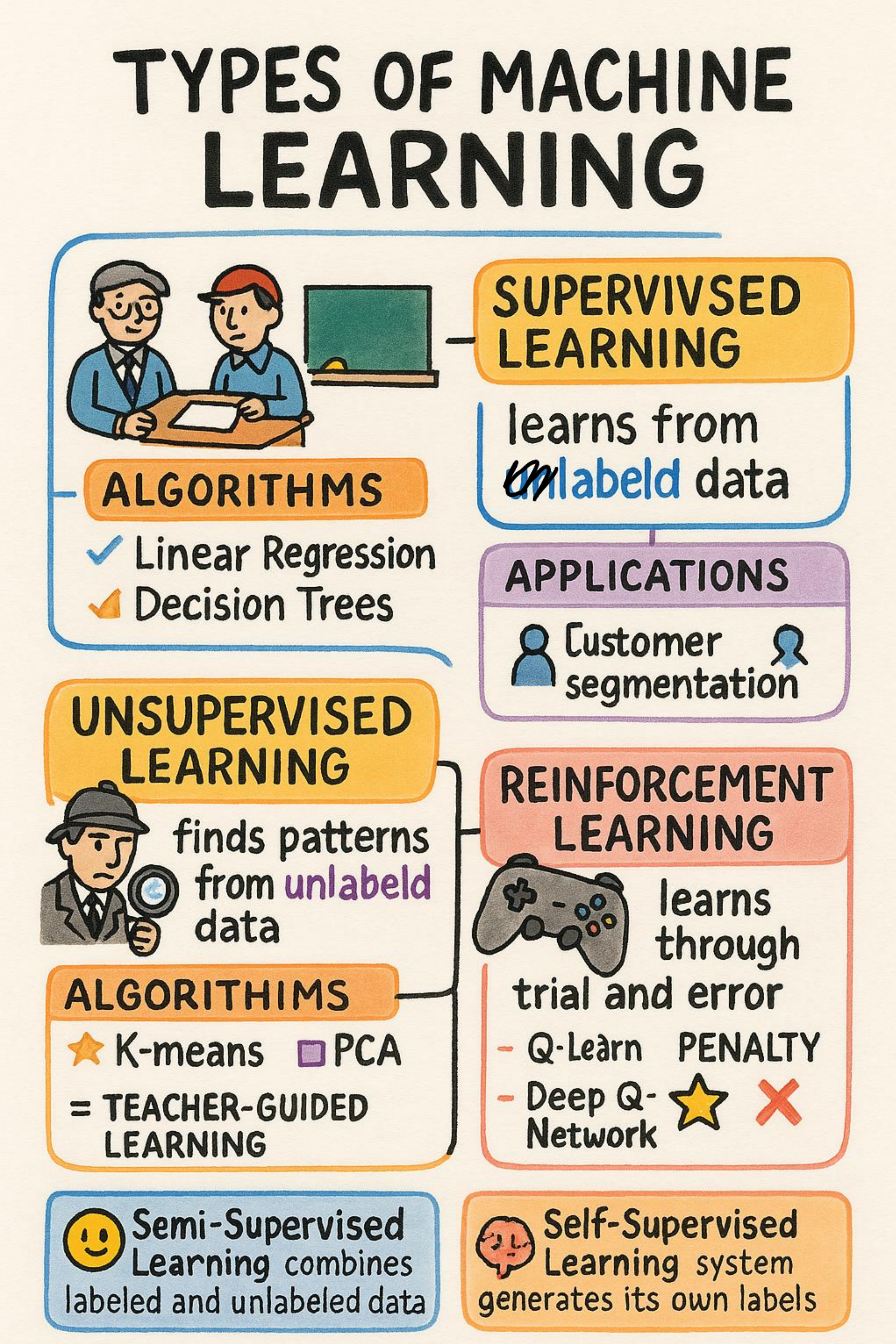
A. Everyday Applications
- Recommendation Rystems
- Netflix, YouTube, Amazon

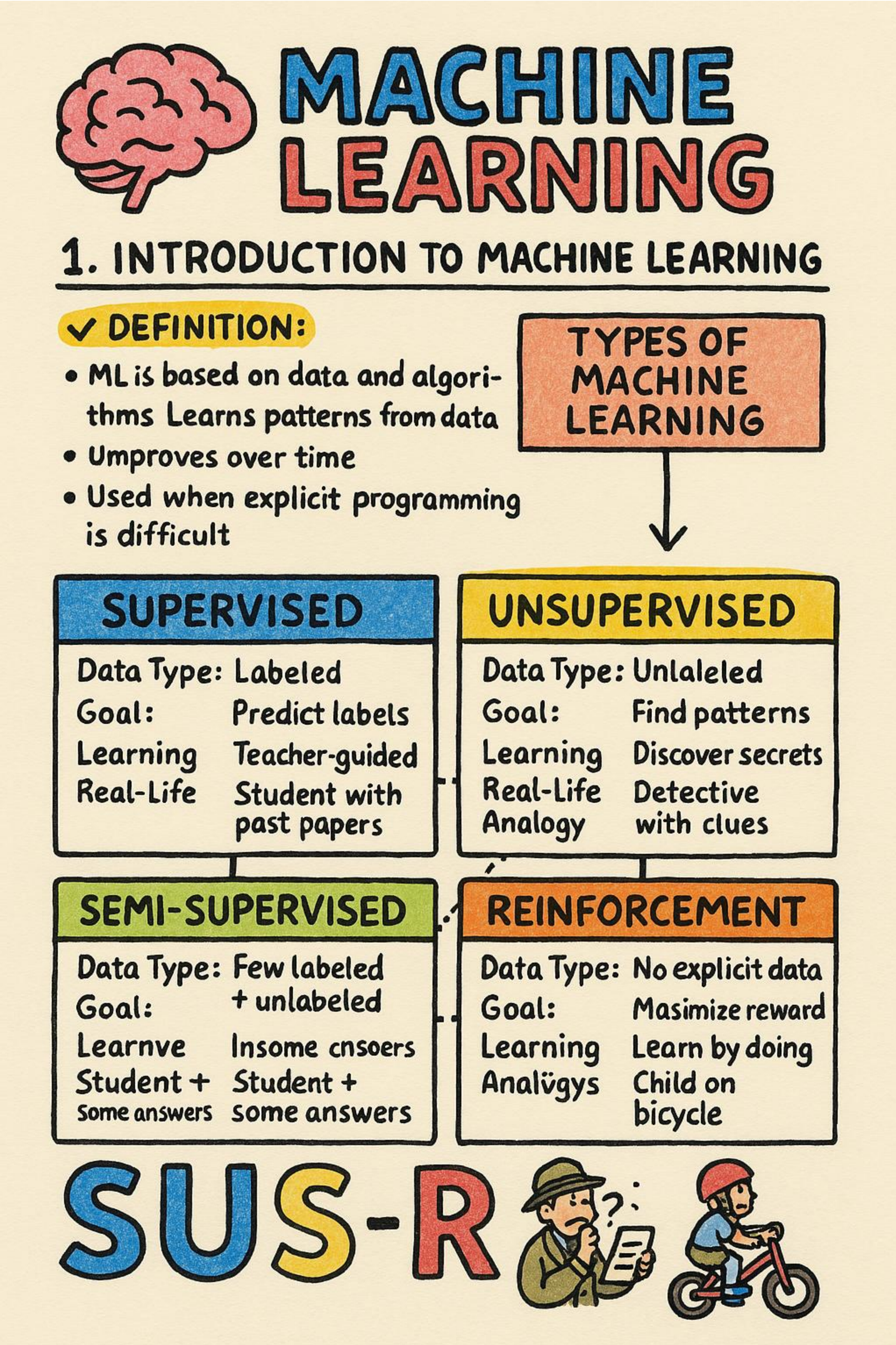
C. Business & Finance
- Credit Scoring & Froud Detection
- Stack Price Prediction

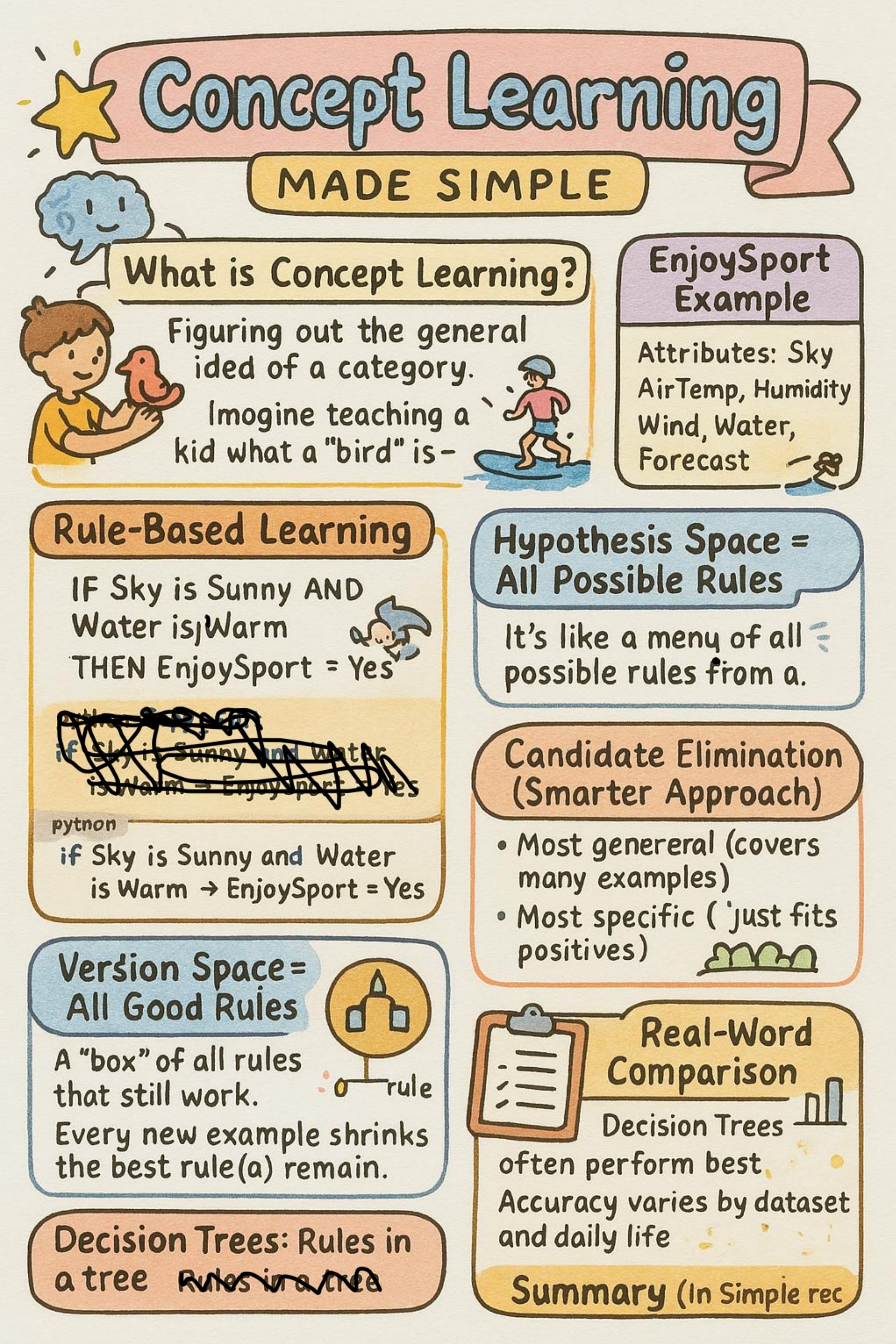
B. Autonomous Systems
- Self-driving cars
- Robotics navigation

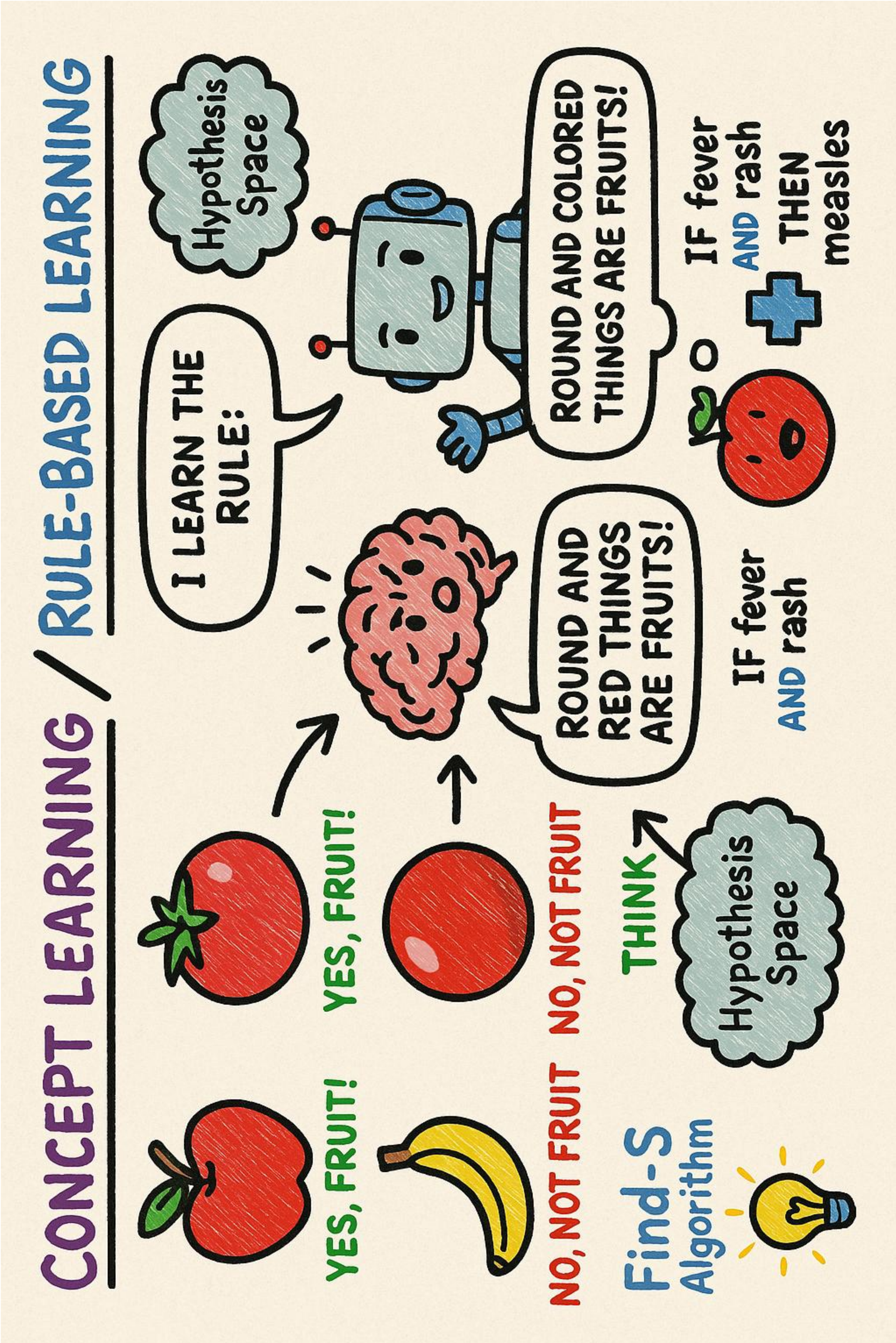
E. Security
- Face recognition
- Tatnitine

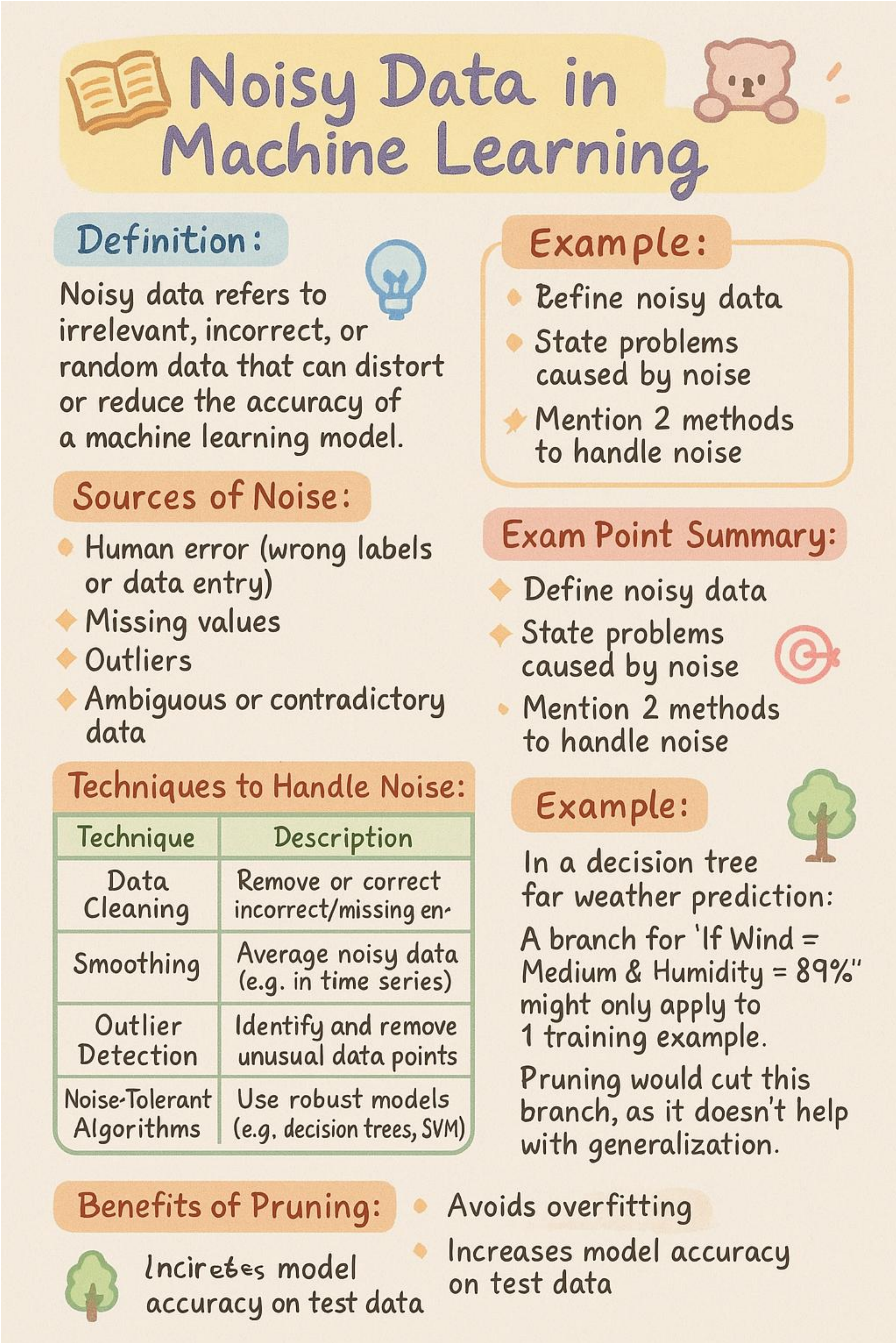
Trick to Memorize:
R H B A S

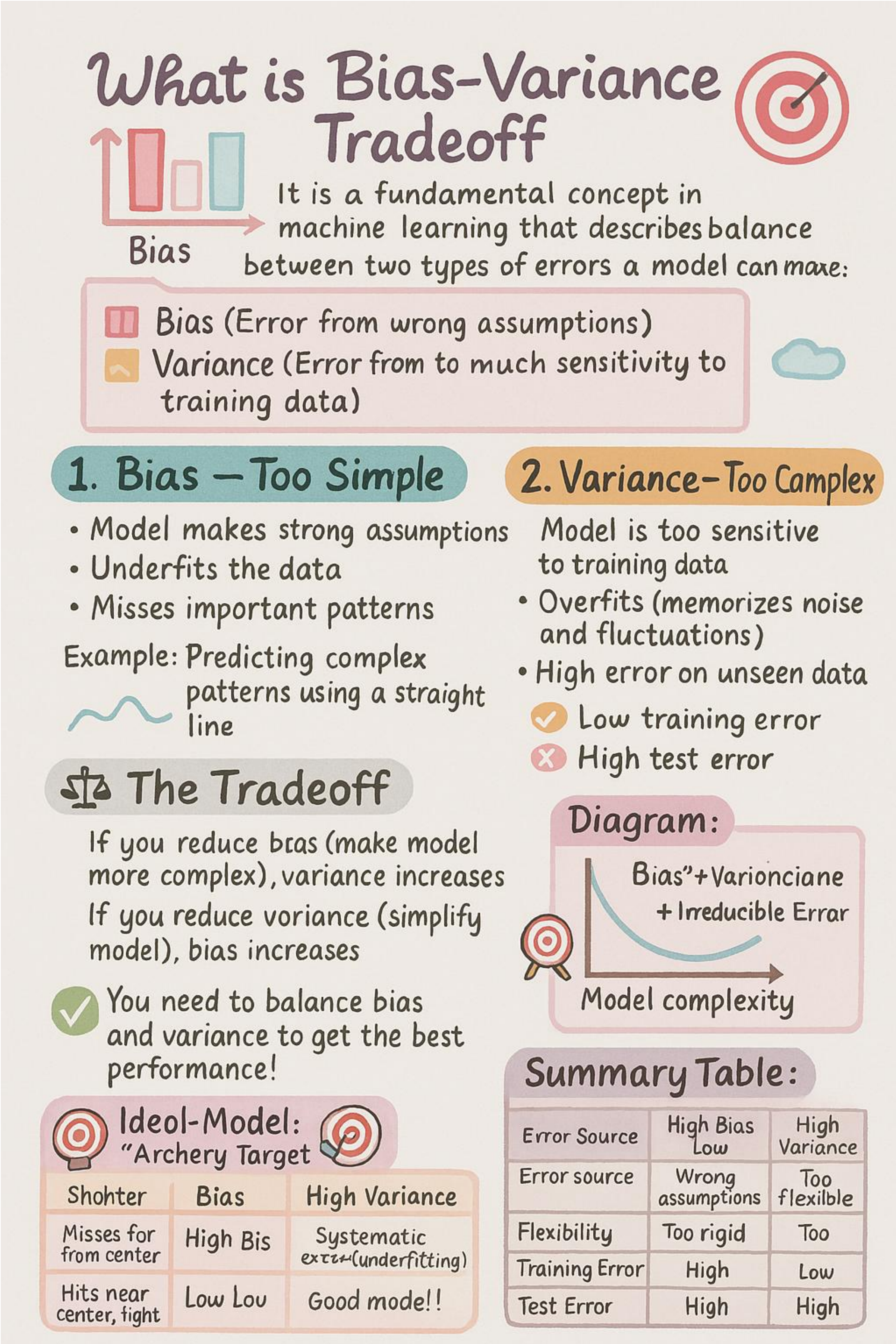

TYPES OF MACHINE LEARNING
SUPERVIVSED LEARNING
learns from ~~un~~labeld data
ALGORITHMS
Linear Regression
Decision Trees
APPLICATIONS
Customer segmentation
UNSUPERVISED LEARNING
finds patterns from unlabeld data
ALGORITHIMS
K-means
PCA
= TEACHER-GUIDED LEARNING
REINFORCEMENT LEARNING
learns through trial and error
- Q-Learn
PENALTY
- Deep Q-Network
Semi-Supervised Learning combines labeled and unlabeled data
Self-Supervised Learning system generates its own labels

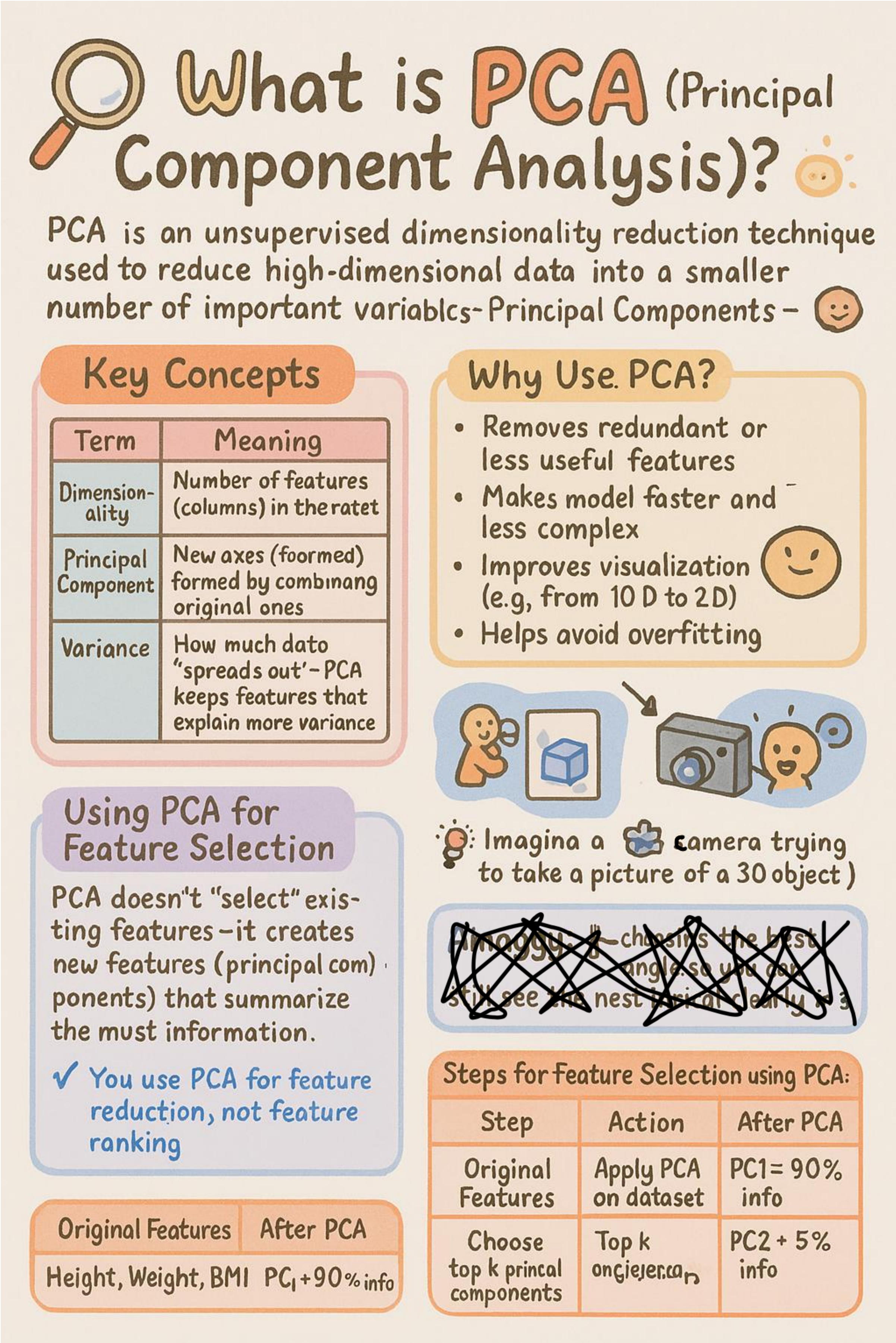

MACHINE LEARNING
1. INTRODUCTION TO MACHINE LEARNING
DEFINITION:
• ML is based on data and algorithms Learns patterns from data
• Umproves over time
• Used when explicit programming is difficult
TYPES OF MACHINE LEARNING
SUPERVISED
Data Type: Labeled
Goal: Predict labels
Learning Teacher-guided
Real-Life Student with past papers
UNSUPERVISED
Data Type: Unlaleled
Goal: Find patterns
Learning Discover secrets
Real-Life Analogy Detective with clues
SEMI-SUPERVISED
Data Type: Few labeled + unlabeled
Goal:
Learnve Insome cnsoers
Student + some answers Student + some answers
REINFORCEMENT
Data Type: No explicit data
Goal: Masimize reward
Learning Learn by doing
Analvgys Child on bicycle
SUS-R

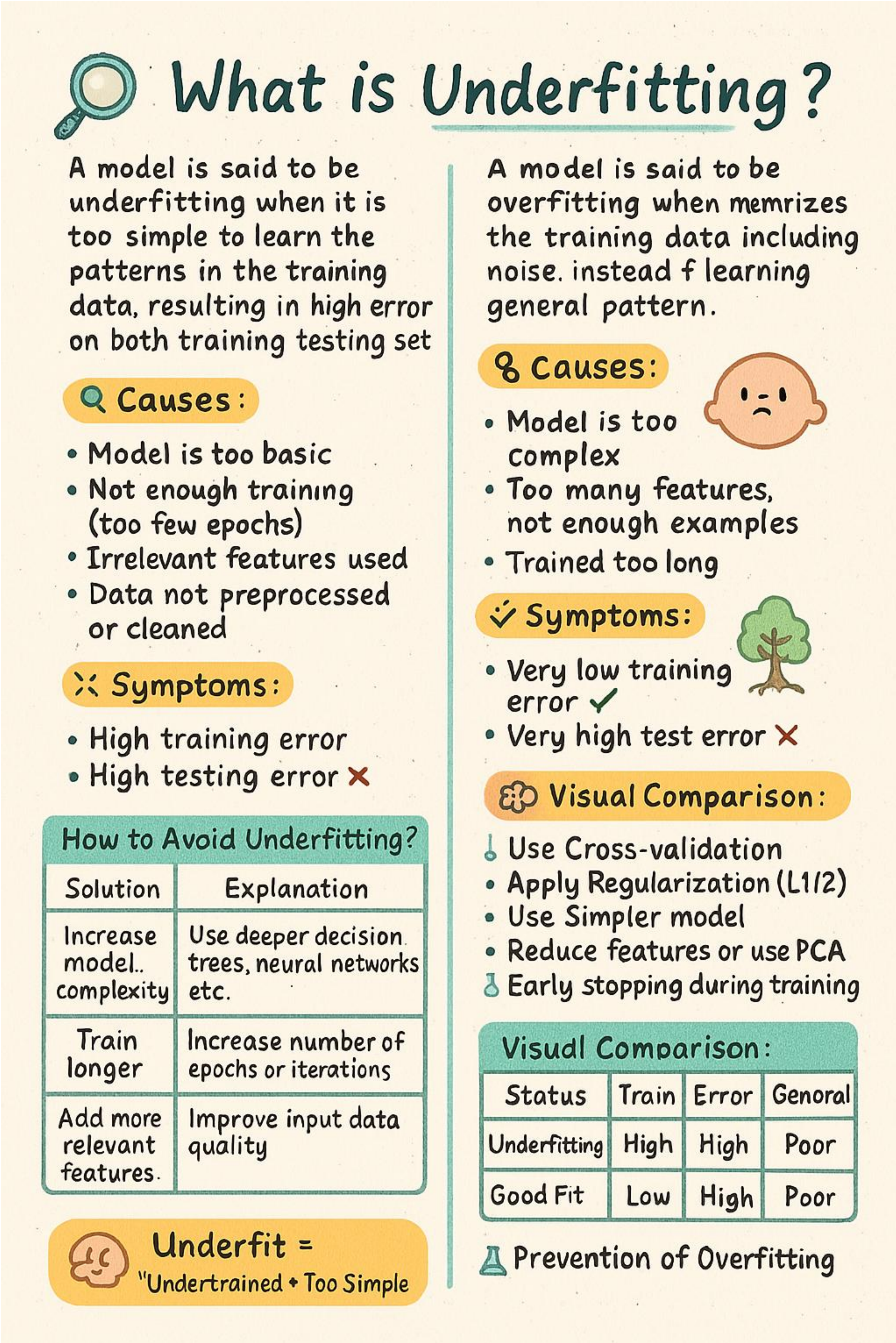

Concept Learning
MADE SIMPLE
What is Concept Learning?
Figuring out the general ided of a category.
Imogine teaching a kid what a "bird" is-
EnjoySport Example
Attributes: Sky AirTemp, Humidity Wind, Water, Forecast
Rule-Based Learning
IF Sky is Sunny AND Water is Warm THEN EnjoySport = Yes
pytnon
if Sky is Sunny and Water is Warm → EnjoySport = Yes
Hypothesis Space = All Possible Rules
It's like a menu of all possible rules from a.
Candidate Elimination (Smarter Approach)
• Most genereral (covers many examples)
• Most specific ('just fits positives)
Version Space = All Good Rules
A "box" of all rules that still work.
rule
Every new example shrinks the best rule(a) remain.
Decision Trees: Rules in a tree
Real-Word Comparison
Decision Trees often perform best
Accuracy varies by dataset and daily life
Summary (In Simple rec


CONCEPT LEARNING / RULE-BASED LEARNING
YES, FRUIT!
YES, FRUIT!
NO, NOT FRUIT
NO, NOT FRUIT
I LEARN THE RULE:
Hypothesis Space
ROUND AND RED THINGS ARE FRUITS!
ROUND AND COLORED THINGS ARE FRUITS!
Find-S Algorithm
THINK
Hypothesis Space
IF fever AND rash
IF fever AND rash THEN measles


Noisy Data in Machine Learning
Definition:
Noisy data refers to irrelevant, incorrect, or random data that can distort or reduce the accuracy of a machine learning model.
Example:
Refine noisy data
State problems caused by noise
Mention 2 methods to handle noise
Sources of Noise:
Human error (wrong labels or data entry)
Missing values
Outliers
Ambiguous or contradictory data
Exam Point Summary:
Define noisy data
State problems caused by noise
Mention 2 methods to handle noise
Techniques to Handle Noise:
Technique
Description
Data Cleaning
Remove or correct incorrect/missing en-
Smoothing
Average noisy data (e.g. in time series)
Outlier Detection
Identify and remove unusual data points
Noise-Tolerant Algorithms
Use robust models (e.g. decision trees, SVM)
Example:
In a decision tree far weather prediction:
A branch for 'If Wind = Medium & Humidity = 89%"
might only apply to 1 training example.
Pruning would cut this branch, as it doesn't help with generalization.
Benefits of Pruning:
Avoids overfitting
Increases model accuracy on test data
Inciretes model accuracy on test data


What is Bias-Variance Tradeoff
Bias
It is a fundamental concept in machine learning that describes balance between two types of errors a model can make:
Bias (Error from wrong assumptions)
Variance (Error from to much sensitivity to training data)
1. Bias – Too Simple
• Model makes strong assumptions
• Underfits the data
• Misses important patterns
Example: Predicting complex patterns using a straight line
2. Variance– Too Complex
Model is too sensitive to training data
• Overfits (memorizes noise and fluctuations)
• High error on unseen data
Low training error
High test error
The Tradeoff
If you reduce bias (make model more complex), variance increases
If you reduce voriance (simplify model), bias increases
You need to balance bias and variance to get the best performance!
Diagram:
Bias"+Varioncione + Irreducible Errar
Model complexity
Ideol-Model: "Archery Target
Shohter
Bias
High Variance
Misses for from center
High Bis
Systematic (underfitting)
Hits near center, tight
Low Lou
Good mode!!
Summary Table:
Error Source
High Bias Low
High Variance
Error source
Wrong assumptions
Too flexible
Flexibility
Too rigid
Too
Training Error
High
Low
Test Error
High
High


What is PCA (Principal Component Analysis)?
PCA is an unsupervised dimensionality reduction technique used to reduce high-dimensional data into a smaller number of important variablcs- Principal Components -
Key Concepts
Term
Meaning
Dimension-ality
Number of features (columns) in the ratet
Principal Component
New axes (foormed) formed by combinang original ones
Variance
How much dato "spreads out'- PCA keeps features that explain more variance
Why Use PCA?
• Removes redundant or less useful features
• Makes model faster and less complex
• Improves visualization (e.g, from 10 D to 2D)
• Helps avoid overfitting
Imagina a camera trying to take a picture of a 30 object)
Using PCA for Feature Selection
PCA doesn't "select" existing features - it creates new features (principal com) ponents) that summarize the must information.
✓ You use PCA for feature reduction, not feature ranking
Original Features
After PCA
Height, Weight, BMI
PC1 + 90% info
Steps for Feature Selection using PCA:
Step
Action
After PCA
Original Features
Apply PCA on dataset
PC1 = 90% info
Choose top k princal components
Top k oncjejer.can
PC2 + 5% info


What is Underfitting?
A model is said to be underfitting when it is too simple to learn the patterns in the training data, resulting in high error on both training testing set
Causes:
• Model is too basic
• Not enough training (too few epochs)
• Irrelevant features used
• Data not preprocessed or cleaned
Symptoms:
• High training error
• High testing error ✗
How to Avoid Underfitting?
Solution | Explanation
Increase model.. complexity | Use deeper decision trees, neural networks etc.
Train longer | Increase number of epochs or iterations
Add more relevant features. | Improve input data quality
Underfit =
"Undertrained + Too Simple
A model is said to be overfitting when memrizes the training data including noise. instead f learning general pattern.
Causes:
• Model is too complex
• Too many features, not enough examples
• Trained too long
Symptoms:
• Very low training error ✓
• Very high test error ✗
Visual Comparison:
Use Cross-validation
• Apply Regularization (L1/2)
• Use Simpler model
• Reduce features or use PCA
Early stopping during training
Visudl Comparison:
Status | Train | Error | Genoral
Underfitting | High | High | Poor
Good Fit | Low | High | Poor
Prevention of Overfitting

LINEAR REGRESSION

KEY IDEA:

Linear Regression is used to predict continuous values (e.g., price, height, marks).
It finds best straight line (regresssion line) that fits te data points

FORMULA:

$y = mx + c$ or $w_1 x w_0$

Variable	Description
y	Predicted output
x	Input feature
m or w_1	Slope/weight
c or w_0	Bias/intercept

EXAMPLE

- Is email spam or not?
- Will a student pass or fail?
- Is a patient diabetic or not?

SIGMOID FUNCTION

$$P(y=1|x) = \frac{1}{1+e^{-xw_0}}$$

Variable	Description
$P(y=1|x)$	Probability of class 1

✓ USES:

- Binary classification (2 classes)
- Extended to multi-class classification (using Softmax)

DIFFERENCE BETWEEN LINEAR & LOGISTIC REGRESSION

Feture	Linear	Logistic Regression	
Output Type	Continuous (Real numbers)	Categorical (Probabilities	Probabillties or 0/1)
Butput Range	-∞ to +∞	Between	0 and 1
Function Used	Linear (straight line)	Sigmoid (S-curve)	Classification tasks

is SVM
(Support Vector Machine)

Key Concepts

Term	Meaning
Hyperplane	A line (in 20), plane (in 30), or higher dimensional surface
Support Vectors	The data point closest to the hyperplane
Margin	The distance in between hyperplane and support vectors
Maximum Margin	SM aims to maximize fidistance for better generalization

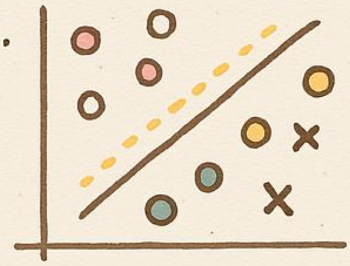
Why Use SVM?

- Works well in high-dimensi-spaces
- Effective when data is not linearly seporable (uses kernel trick)

Types of SVM

Used when	Data can dime
Non-linear SVM	Toecpaile catroed equitannes

SVM in 2D
(Simple Visualization Idea)

1.


1. Find the widest possible gap between
2. Keep the line (hyperplane) in the middle
3. Chclosest dots en each-side = support vectors

Kernel Trick

Converts non-linearly separubile data into higher dimension

Good for text classification, image recognition bioinformatics

Applications of SVM:

✓ Spam detection
✓ Handwriting recognition
✓ Face detection

What is a Decision Tree?

A Decision Tree is a supervised learning algorithm used for classification and regression. It splits the data into branches based on fecture values until it reaches a decision (leaf node).

Simple Definition:

- A tree-like structure where:
- Each internal node = a test on a feature
- Each branch = a result of test
- Each leaf node = a class label (decision)

Real-Life Example:

Should I go out?

Weather?

Sunny → Play

Rainy → Stay home

Types of Decision Trees

Type	Used For
Classification Tree	Classification
Regression Tree	Continuous Valve (e.g. salary)

Advantages

- Easy to understand and visualize

Key Terms:

Term	Meaning
Root Node	First feature spllt
Internal Node	A question/test on feature (e.g. Age ≥50?)
Leaf Node	Final decision/output (e.g. Yes/No, Class A/B)
Branch	Outcome of the test (True/False path)
Entrapy	Measures imparity or randomness in the data
Gini Index	Another impurity measure (used in CART algorithm)

✓ Advantages

- Can to understand and visualize
- No need for feature scaling
- Works for both numeric and categorical data
- Fast on small to medium datasets

Trick to Avoid Overfitting in Decision Trees:

- Use Pruning (cut branches that don't improve accuracy)
- Set max depth or min samjles per leaf